MILITARY SPECIFICATION SHEET

WRENCH, BOX, RATCHETING, SPLINED, HIGH STRENGTH, THIN WALL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the ratcheting box wrenches described herein shall consist of this document and the latest issue of MIL-W-8982.

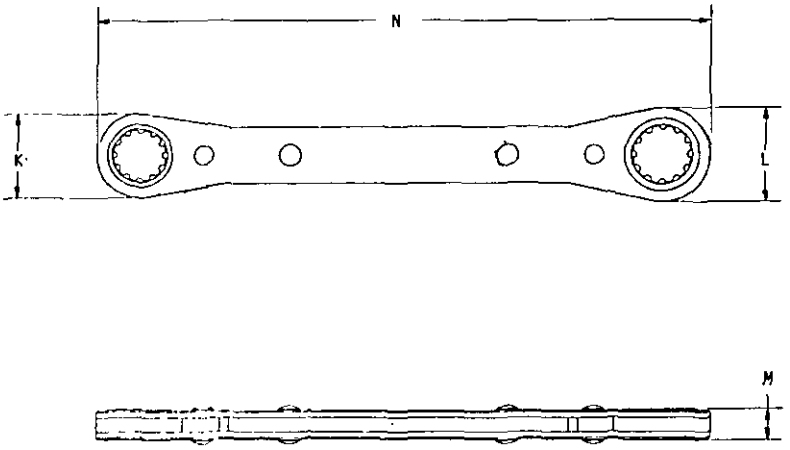


FIGURE 1. Ratcheting Spline Box wrench.

DESIGN REQUIREMENTS:

1. The ratcheting box wrench shall conform to figure 1 and the dimensions in table I. Spline and chamfer dimensions shall conform to MS33787. This wrenching design fits fasteners with the corresponding MS33787 spline as well as *bihexagonal and hexagonal wrenching configurations.*

2. The gear wheel (socket) and pawl shall be hardened to 50-54 HRC and 52-56 HRC respectively.

AMSC N/A FSC 5120

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3. The top and bottom plates shall be chromium plated in accordance with QQ-C-320. The remaining components shall be either black oxide plated or manganese phosphate plated (which shall consist of a chemically produced oxide or manganese phosphate, followed with a coating of rust-preventative oil) or chromium plated.

PERFORMANCE REQUIREMENTS:

1. The wrench shall withstand the endurance torque load and ultimate torque load values specified in table I. There shall be no damage or permanent deformation (set) to the wrench.

2. The endurance torque and ultimate torque loads shall be applied by any suitable manner or mechanical means. Measures should be taken to prevent slippage of wrenches on mandrels during tests.

3. When engaged with hexagon and spline mandrels, the wrench shall be tested for conformance to the endurance torque load and ultimate torque load values specified in table I. The load shall be held on each for 10 seconds. The size of the hexagon mandrel in the engaged portion shall be equal to (+ 0.000, -0.002) the basic size of the wrench specified. Spline mandrel dimensions in the engaged portions shall be in accordance with MS33787, table I, except that C and W dimensions shall be held to the minimum dimension (+ 0.002, - 0.000). Mandrels shall be hardened to not less than 55 HRC and smoothly finished in the engagement area of wrench openings.

4. The wrench shall be subjected to not less than 500 torque cycles on both the hexagon and spline mandrels, in accordance with the endurance torque load and ultimate torque load values specified in table I.

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TABLE I. Ratcheting box wrench dimensions and test loads.

Combination Number	Wrench Opening Spline Number		Basic Wr. Size		Outside Dia. of Head (Max.)		M Thickness of each Head (Max.) (In.)	N Overall Length (Min.) (In.)	Spline & Single Hex Tests			
									Small Head		Large Head	
	Small Head	Large Head	Small Head (In.)	Large Head (In.)	K Small Head (In.)	L Large Head (In.)			Ultimate Torque (Lb. In.)	Endurance Torque (Lb. In.)	Ultimate Torque (Lb. In.)	Endurance Torque (Lb. In.)
01	8	10	.250	.312	.750	.813	.438	3.75	144	100	180	126
02	12	14	.375	.438	.938	1.063	.469	5.25	360	252	420	294
03	16	18	.500	.562	1.156	1.250	.500	6.50	600	420	720	504
04	18	20	.562	.625	1.250	1.375	.500	7.75	720	504	900	630
05	20	22	.625	.688	1.375	1.562	.500	7.75	900	630	930	651
06	20	24	.625	.750	1.375	1.562	.500	7.75	900	630	960	672
07	22	28	.688	.875	1.562	1.562	.500	8.62	930	651	1,060	742
08	24	28	.750	.875	1.562	1.562	.500	8.62	960	672	1,060	742
09	26	30	.813	.938	1.562	1.625	.750	8.62	1,000	700	1,100	770

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NOTES:

1. Dimensions are in inches.

2. The wrench shall be of uniform quality. The wrench shall be free from nicks, cracks, laps, seams and scale. The wrench shall have no imperfections that will adversely affect serviceability, reliability or safety.

3. Example of a military part number:

M8982/4 - 01

- 01 — Combination number (Small head = .250, large head = .312)
- M8982/4 — Basic specification sheet number

Custodians:
Army - AV
Air Force - 11
Navy - AS

Review:
Army - GL, AR
Air Force - 99

Preparing Activity:
Navy - AS

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